

CHURNGUARD AI

Customer Churn Prediction & Retention Intelligence System

Domain: Artificial Intelligence & Machine Learning

Duration: 2 Days

Language: Python

Environment: Google Colab

1. Problem Statement

Businesses can lose customers after they have decided to leave. CHURNGUARD AI uses historical telecom customer information to predict whether a customer is likely to churn and converts the prediction probability into a simple risk category.

2. Objective

Build an end-to-end supervised classification workflow covering data understanding, cleaning, feature preparation, model training, evaluation, prediction and business interpretation.

3. Dataset Description

The project uses the IBM Telco Customer Churn public sample dataset. The dataset contains 7,043 customer records and 21 original columns. The target variable is Churn, with Yes/No classes.

4. Data Preprocessing

TotalCharges was converted from text to numeric values. Duplicate records were checked and removed where applicable. customerID was excluded from model features. Missing values were handled inside the scikit-learn preprocessing pipeline. Numerical features were imputed/scaled and categorical features were imputed and one-hot encoded.

5. Exploratory Data Analysis

The notebook includes churn distribution, churn by contract, tenure distribution by churn, and monthly-charge distribution by churn. These visualizations are used to identify observed patterns; they do not establish causal relationships.

6. Models Tested

Three model configurations were evaluated: Logistic Regression, Random Forest, and Class-weighted Logistic Regression. The class-weighted model was used as the Day 2 improvement to give greater training weight to the minority churn class.

7. Model Evaluation

Accuracy, Precision, Recall, F1-score and ROC-AUC were calculated on the held-out test set. F1-score was used as the primary selection criterion, followed by Recall and ROC-AUC.

8. Evaluation Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	80.41%	65.72%	55.88%	60.40%	84.19%

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	79.06%	60.48%	47.06%	52.93%	81.87%
Logistic Regression (Balanced)	73.81%	50.43%	78.34%	61.36%	84.13%

Selected model: Logistic Regression (Balanced)

9. Day 2 Improvement

Before: Standard Logistic Regression F1 = 60.40%.

After: Class-weighted Logistic Regression F1 = 61.36%.

What changed? The churn minority class was given greater training weight using `class_weight='balanced'`.

Observed effect: Recall increased from 55.88% to 78.34%, while precision decreased from 65.72% to 50.43%. The F1-score increased from 60.40% to 61.36%. This is a precision-recall trade-off and is relevant when missing actual churners is costly.

10. Final Model Performance

Accuracy: 73.81%

Precision: 50.43%

Recall: 78.34%

F1 Score: 61.36%

ROC-AUC: 84.13%

11. Business Interpretation

CHURNGUARD AI can help a company prioritize customers whose observed characteristics are associated with higher churn probability. High-risk customers can be considered for retention campaigns, service reviews or targeted offers. The model should support human business decisions rather than automatically determining customer treatment.

12. Customer Risk Output

The notebook provides a customer prediction function that returns churn probability and a risk category. Demonstration bands are HIGH for probability $\geq 70\%$, MEDIUM for 40%–69%, and LOW below 40%. These thresholds are project demonstration thresholds and should be calibrated before production use.

13. Important Feature Signals

The final model output identified strong predictive signals including tenure, contract type, internet service and monthly charges. These are predictive signals in the trained model, not proof that any individual feature causes churn.

14. Limitations

Predictions are probabilistic and are not guarantees. The dataset is historical and may not represent future customer behavior. The observed relationships do not establish causation. Production deployment would require validation on future data, probability calibration, threshold optimization and model monitoring.

15. Future Improvements

Future work can include cross-validated hyperparameter tuning, temporal validation, probability calibration, cost-sensitive threshold optimization, SHAP-based explainability, Streamlit deployment and model/data-drift monitoring.

16. Final Results

Dataset records: 7,043

Models tested: Logistic Regression, Random Forest, Logistic Regression (Balanced)

Selected model: Logistic Regression (Balanced)

Accuracy: 73.81%

Precision: 50.43%

Recall: 78.34%

F1 Score: 61.36%

ROC-AUC: 84.13%